How to Use PVTP

Good day to my fellow young engineers.

I know the move from classroom to work desk is very hectic. It's harder to understand things, and the skills you have always used to excel can seem somewhat subpar now.

Hang on. You are not alone, and you are doing fine.

This document is specifically for those in oil and gas. I created it to explain how to use the PVTP software for PVT analysis. I hope it makes someone's journey easier.

Thanks.

HOW TO USE PVTP

BY ANNIE O.

DISCLAIMER

I am not a professional, I just like learning and sharing.

I used the WF OIL REC sample CL-70055, which I got from Whitson.com

https://manual.pvt.whitson.com/certificate/data_digitalization/pvt_experiment_data/

Software used is PVTP by PE(Petroleum Experts).



WHAT IS PVT?

PVT stands for Pressure, Volume and Temperature. It refers to the laboratory analysis and experiments used to characterize reservoir fluids and to understand how their properties and phase behaviour change with pressure and temperature. PVT analysis helps us understand how the fluid will behave under reservoir and surface conditions as it is produced from the reservoir to the surface.

WHY DO WE PERFORM PVT ANALYSIS?

- To understand how the reservoir fluid behaves and how its properties change under different pressure and temperature conditions.
- To characterize the fluid and develop a reliable PVT model that can be used for material-balance calculations, reservoir simulation and dynamic modelling.
- To provide the fluid-property and phase-behaviour information required for the design and operation of surface facilities, such as separators and other processing equipment.

DATA OBTAINED FROM PVT ANALYSIS

1. Compositional Analysis

Compositional analysis determines the chemical composition of the reservoir fluid by identifying the individual components and heavier hydrocarbon fractions present. It shows us how much of each pure hydrocarbon (C1, C2, C3), non-hydrocarbon (nitrogen, carbon dioxide) and pseudo-component is in the fluid. Pseudo-components are heavy fractions grouped together and written with a plus sign after them, e.g. C7+. The heavy fractions are characterized using properties such as molecular weight, density/specific gravity and other properties required for fluid modelling.

2. Constant Composition Expansion (CCE)

The Constant Composition Expansion experiment is used to study how fluid volume and phase behaviour change as pressure changes while the overall fluid composition stays constant.

No fluid is removed from the cell. For an oil system, as pressure is reduced below the bubble point, gas evolves from the oil, but the liberated gas stays inside the cell. Important information from this experiment includes the bubble-point pressure, relative volume and phase behaviour.

3. Differential Liberation Experiment (DLE)

The Differential Liberation Experiment is primarily used for reservoir oils. It is used to determine how the oil and the liberated gas behave as pressure is reduced below the bubble point.

At each pressure stage, the gas that evolves from the oil is removed from the cell, and the volume of the remaining oil is measured. Measurements from this experiment give us properties such as the oil formation volume factor (Bo), the solution gas-oil ratio (Rs), oil density and oil viscosity.

4. Constant Volume Depletion (CVD)

The Constant Volume Depletion experiment is primarily used for gas-condensate and other retrograde systems.

Pressure is reduced in stages while the cell volume is kept constant. Below the dew point, liquid condensate forms. At each stage, gas is withdrawn from the cell to bring the volume back to its original value. Measurements such as condensate dropout, produced-gas composition and the properties of the withdrawn gas can be obtained.

AIM OF PVTP

The main aim of PVTP is to develop and tune a fluid model that adequately reproduces the experimentally measured PVT data.

We provide the software with information from laboratory experiments, such as fluid composition, PVT measurements and viscosity data. The software uses an equation of state (EOS) and fluid-characterization methods to calculate the expected fluid properties.

The calculated results are then compared with the laboratory measurements. Where significant differences exist, the fluid model can be tuned by adjusting appropriate parameters. This process is repeated until the model gives an acceptable match to the experimental data.

Once the model reproduces the laboratory data, it can be used to predict fluid properties at other pressure and temperature conditions where laboratory measurements are not available. The fluid model can then be used in material-balance calculations, reservoir simulation, production forecasting and surface-facility design.

Okay, let's get right into it.

STEP 1: EXTRACT YOUR DATA

The first thing to do when you get your lab report is to extract the necessary data into an Excel sheet. For this example, I am using the CL-70055 data (I will link it).

INITIAL RESERVOIR CONDITIONS		
Reservoir Pressure	8213 psia	56.63 MPa
Reservoir Temperature:	228.128 F	108.96 C

CONSTANT COMPOSITION EXPANSION @ 228.1 F (109.0 C)		
Saturation Pressure	3946 psia	27.21 MPa
Compressibility @ Reservoir Pressure	3.21202E-05 $psia^{-1}$	4.658647E-03 MPa^{-1}
Compressibility @ Saturation Pressure	5.10102E-05 $psia^{-1}$	7.398422E-03 MPa^{-1}

DIFFERENTIAL LIBERATION @ 228.1 F (109.0 C)		
At Saturation Pressure		
Oil Formation Volume Factor	3.6916 res.bbl/STB	3.6916 res.m^3/m^3
Solution Gas-Oil Ratio	3918.59 scf/STB	697.90 m^3/m^3
Oil Density	0.4615 g/cm^3	461.5 kg/m^3
Oil Viscosity	0.157 cp	0.2 mPa.s
At Ambient Pressure		
Residual Oil Density	0.7427 g/cm^3	742.7 kg/m^3
Residual Oil Viscosity	0.624 cp	0.6 mPa.s
At Tank Conditions		
Residual Oil Density	0.8116 g/cm^3	811.6 kg/m^3
API Gravity	42.85	42.85

SINGLE-STAGE SEPARATOR TEST		
At Saturation Pressure		
Oil Formation Volume Factor	2.7959 res.bbl/STB	2.7959 res.m^3/m^3
Solution Gas-Oil Ratio	2705.18 scf/STB	481.79 m3/m3
At Tank Conditions		
Residual Oil Density	0.7916 g/cm^3	791.6 kg/m^3
API Gravity	47.26	47.26

This is the basic data we need. It includes the reservoir pressure, temperature, bubble point and every other piece of data we may need for this field. For this example, use a depth of 7546 ft.

TABLE 1
ENCANA CORPORATION
WELL ECA HZ WAHIGAN 16-16-064-23W5 – RECOMBINED SAMPLE
RESERVOIR FLUID STUDY
COMPOSITIONAL ANALYSIS OF RESERVOIR FLUID

Boiling Point (C)			Mole Fraction	Mass Fraction	Calculated Properties	
-195.8	Nitrogen	N2	0.0103	0.0057	**Total Sample**	
-78.5	Carbon Dioxide	CO2	0.0061	0.0053		
-60.3	Hydrogen Sulphide	H2S	0.0000	0.0000	Molecular Weight	50.55
-161.7	Methane	C1	0.5222	0.1658		
-88.9	Ethane	C2	0.1330	0.0791		
-42.2	Propane	C3	0.0791	0.0690	**C6+ Fraction**	
-11.7	i-Butane	i-C4	0.0131	0.0150		
-0.6	n-Butane	n-C4	0.0310	0.0357	Molecular Weight	163.79
27.8	i-Pentane	i-C5	0.0099	0.0141	Mole Fraction	0.1829
36.1	n-Pentane	n-C5	0.0125	0.0178	Density (g/cc)	0.8160
36.1 - 68.9	Hexanes	C6	0.0214	0.0364		
68.9 - 98.3	Heptanes	C7	0.0192	0.0381		
98.3 - 125.6	Octanes	C8	0.0209	0.0471	**C7+ Fraction**	
125.6 - 150.6	Nonanes	C9	0.0155	0.0393		
150.6 - 173.9	Decanes	C10	0.0118	0.0331	Molecular Weight	174.51
173.9 - 196.1	Undecanes	C11	0.0104	0.0301	Mole Fraction	0.1608
196.1 - 215	Dodecanes	C12	0.0080	0.0255	Density (g/cc)	0.8265
215 - 235	Tridecanes	C13	0.0079	0.0272		
235 - 252.2	Tetradecanes	C14	0.0063	0.0239		
252.2 - 270.6	Pentadecanes	C15	0.0048	0.0195	**C12+ Fraction**	
270.6 - 287.8	Hexadecanes	C16	0.0039	0.0172		
287.8 - 291.7	Heptadecanes	C17	0.0032	0.0149	Molecular Weight	271.05
291.7 - 317.2	Octadecanes	C18	0.0030	0.0149	Mole Fraction	0.0603
317.2 - 330	Nonadecanes	C19	0.0027	0.0142	Density (g/cc)	0.8756
330 - 344.4	Eicosanes	C20	0.0022	0.0118		
344.4 - 357.2	Heneicosanes	C21	0.0019	0.0108		
357.2 - 369.4	Docosanes	C22	0.0016	0.0098		
369.4 - 380	Tricosanes	C23	0.0015	0.0093		
380 - 391.1	Tetracosanes	C24	0.0013	0.0086		
391.1 - 401.7	Pentacosanes	C25	0.0012	0.0081		
401.7 - 412.2	Hexacosanes	C26	0.0010	0.0072		
412.3 - 422.2	Heptacosanes	C27	0.0009	0.0069		
422.3 - 431.7	Octacosanes	C28	0.0008	0.0061		
431.7 - 441.1	Nonacosanes	C29	0.0007	0.0060		
Above 441.1	Tricontanes Plus	C30+	0.0073	0.0812		
48.9	Cyclopentane	C5H10	0.0007	0.0009		
72.2	Methylcyclopentane	C6H12	0.0034	0.0057		
81.1	Cyclohexane	C6H12	0.0028	0.0046		
101.1	Methylcyclohexane	C7H14	0.0074	0.0143		
80.0	Benzene	C6H6	0.0005	0.0007		
110.6	Toluene	C7H8	0.0019	0.0035		
136.1 - 138.9	Ethylbenzene & p,m-Xylene	C8H10	0.0022	0.0045		
144.4	o-Xylene	C8H10	0.0017	0.0035		
168.9	1, 2, 4-Trimethylbenzene	C9H12	0.0030	0.0071		
Total			**1.0000**	**1.0000**		

This is the compositional analysis for this field. It lists the hydrocarbon components and their corresponding molar percent (mol %), which is what will be input into PVTP. The properties of the pseudo-components are also listed.

The fluid composition comes in two forms, surface and downhole:

- **Surface:** the fluid is sampled at the separator (separator liquid and separator gas) and then recombined to represent the fluid downhole.
- **Downhole:** the fluid is taken using bottom-hole sampling. It represents the fluid at reservoir temperature and pressure.

All the molar percents must add up to 100.

TABLE 3
ENCANA CORPORATION
WELL ECA HZ WAHIGAN 16-16-064-23W5 – RECOMBINED SAMPLE
RESERVOIR FLUID STUDY
CONSTANT COMPOSITION EXPANSION @ 228.1 °F (109.0 °C)

Pressure (psia)	Pressure (MPa)	Relative Volume [1]	Y-Function [2]	Fluid Density (g/cc)
5544	38.22	0.938146		0.4919
5197	35.83	0.948720		0.4864
4884	33.67	0.959445		0.4810
4602	31.73	0.970172		0.4757
4350	29.99	0.980768		0.4705
4126	28.45	0.991014		0.4657
4024	27.74	0.996021		0.4633
3946 Psat	27.21	**1.000000**		**0.4615**
3872	26.69	1.008499	2.2488	
3770	25.99	1.021057	2.2172	
3675	25.34	1.033707	2.1878	
3322	22.90	1.090372	2.0786	
3021	20.83	1.154221	1.9855	
2793	19.26	1.215586	1.9150	
2447	16.87	1.338851	1.8080	
2195	15.13	1.461143	1.7301	
1998	13.77	1.584190	1.6691	
1705	11.75	1.832782	1.5785	
1494	10.30	2.084750	1.5132	
1335	9.20	2.336109	1.4641	
1203	8.29	2.602390	1.4232	
1101	7.59	2.857146	1.3917	
1015	7.00	3.115874	1.3651	

[1] Volume at indicated pressure per volume at saturation pressure
[2] Y Function = ((Psat-P)/P)/(Relative Volume - 1)
Psat - Saturation Pressure

This is the CCE data. The two columns we use for this example are pressure (psia) and relative volume. The main property we get from this experiment is the bubble point.

TABLE 4
ENCANA CORPORATION
WELL ECA HZ WAHIGAN 16-16-064-23W5 – RECOMBINED SAMPLE
RESERVOIR FLUID STUDY
DIFFERENTIAL LIBERATION OIL PROPERTIES @ 228.1°F (109.0°C)

Pressure (psia)	Pressure (MPa)	Oil Density (g/cm^3)	Oil Formation Volume Factor [1]	Total Formation Volume Factor [2]	Gas-Oil Ratio Solution (scf/STB)	Gas-Oil Ratio Liberated (scf/STB)	Gas-Oil Ratio Solution (m^3/m^3)	Gas-Oil Ratio Liberated (m^3/m^3)
5544	38.22	0.4919	3.4632	3.4632	3918.59	0.00	697.90	0.00
5197	35.83	0.4864	3.5023	3.5023	3918.59	0.00	697.90	0.00
4884	33.67	0.4810	3.5419	3.5419	3918.59	0.00	697.90	0.00
4602	31.73	0.4757	3.5815	3.5815	3918.59	0.00	697.90	0.00
4350	29.99	0.4705	3.6206	3.6206	3918.59	0.00	697.90	0.00
4126	28.45	0.4657	3.6584	3.6584	3918.59	0.00	697.90	0.00
4024	27.74	0.4633	3.6769	3.6769	3918.59	0.00	697.90	0.00
3946 Psat	27.21	**0.4615**	**3.6916**	**3.6916**	**3918.59**	**0.00**	**697.90**	**0.00**
3513	24.22	0.4981	2.8739	3.8480	2718.48	1200.12	484.16	213.74
3113	21.46	0.5264	2.5265	4.1148	2159.22	1759.38	384.56	313.35
2713	18.71	0.5512	2.2764	4.4438	1816.32	2102.28	323.49	374.42
2313	15.95	0.5733	2.1122	4.9987	1537.08	2381.51	273.76	424.15
1913	13.19	0.5945	1.9743	5.8521	1319.40	2599.20	234.99	462.92
1513	10.43	0.6146	1.8317	7.2971	1085.44	2833.15	193.32	504.59
1113	7.67	0.6369	1.7196	9.8887	879.39	3039.21	156.62	541.29
713	4.92	0.6841	1.6249	15.4569	713.09	3205.51	127.00	570.90
313	2.16	0.6829	1.4757	35.4321	500.45	3418.14	89.13	608.77
13	0.09	0.7427	1.0842	525.4128	0.00	3918.59	0.00	697.90

Density of Residual Oil = 0.8116 g/cm3 (811.6 kg/m3) @ 60 F (288.7K)

[1] Barrels (Cubic meters) of oil at indicated pressure and temperature per barrel (cubic meter) of residual oil @ 60 F (288.7 K).
[2] Total barrels (cubic meters) of oil and liberated gas at the indicated pressure and temperature per barrel (cubic meter) of residual oil @ 60 F (288.7 K).
Psat - Saturation Pressure
- Tank conditions: 60 F (288.7 K) @ 13 psia (0.0896 MPa); Standard conditions: 60 F (288.7 K) @ 14.696 psia (0.101325 MPa).

This is the differential liberation experiment. The columns we use are pressure (psia), oil density, oil formation volume factor (Bo) and solution gas-oil ratio (Rs, scf/stb).

TABLE 6
ENCANA CORPORATION
WELL ECA HZ WAHIGAN 16-16-064-23W5 – RECOMBINED SAMPLE
RESERVOIR FLUID STUDY
DIFFERENTIAL LIBERATION FLUID VISCOSITY @ 228.1°F (109.0°C)

Pressure (psia)	Pressure (MPa)	Oil Viscosity (cp=mPa.s)	Gas Viscosity (cp=mPa.s)	Oil - Gas Viscosity Ratio
4713	32.49	0.167		
4413	30.43	0.164		
4113	28.36	0.160		
3946 Psat	27.21	0.157		
3513	24.22	0.163	0.03068	5.33
3113	21.46	0.178	0.02563	6.95
2713	18.70	0.190	0.02216	8.58
2313	15.95	0.204	0.01966	10.39
1913	13.19	0.223	0.01762	12.65
1513	10.43	0.250	0.01612	15.51
1113	7.67	0.288	0.01494	19.25
713	4.91	0.334	0.01392	24.02
313	2.16	0.420	0.01250	33.61
13	0.09	0.624	0.01010	61.73

Psat - Saturation Pressure

The last table is the oil viscosity table. Sometimes viscosity is included in the DL table. On occasions like this, where it has its own table, we must harmonize the pressures and interpolate or extrapolate the missing viscosity values. The column we use is oil viscosity. Viscosity is usually regressed and calculated separately in the PVT software.

STEP 2: QA/QC TEST

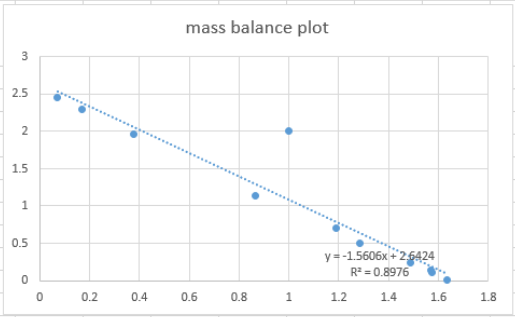
Next, we test the integrity of our data.

This data (CL-70055) does not need a QA/QC test because the composition came from a downhole sample.

If it were surface data, we would create two new columns, xi/zi and yi/zi, and plot yi/zi against xi/zi. Here x is the separator liquid composition, y is the separator gas composition and z is the recombined composition. Then we find the R^2 value. The closer it is to 1, the more consistent our data is.

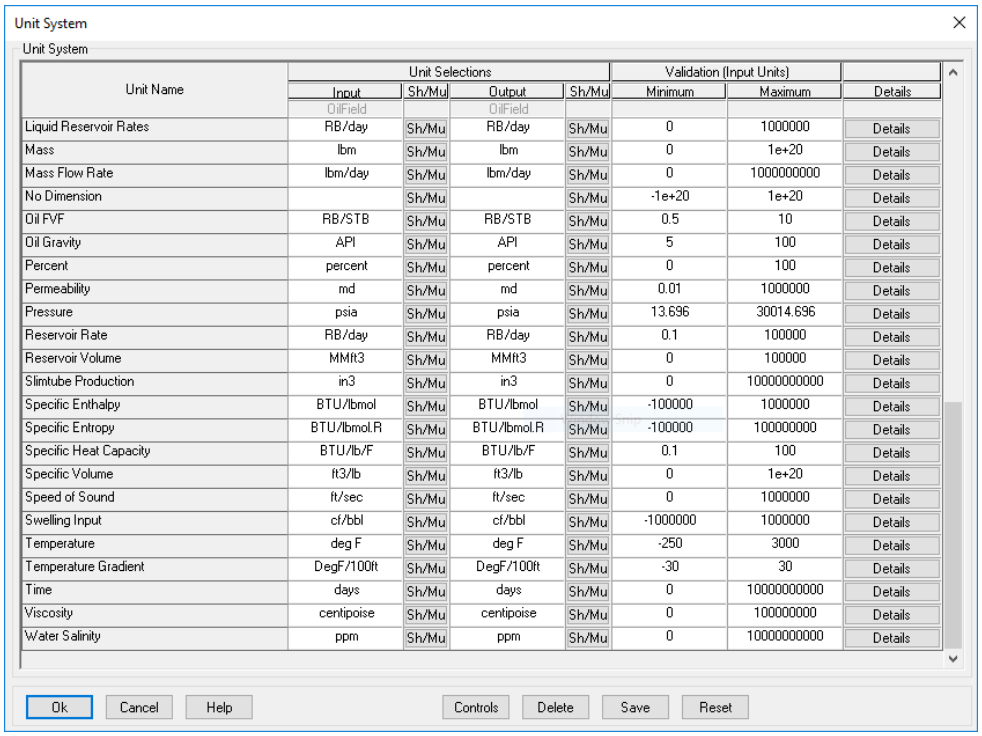
Example (this is not CL-70055, it is just another example to explain the QA/QC test):

Fluid Composition					
Component	Seperator Liquid (X)	Seperator Gas (Y)	Reservoir Fluid (Z)	Yi/Zi	Xi/Zi
N2	0.01	0.02	0.01	2	1
CO2	0.54	7.33	3.19	2.297806	0.169279
C1	2.54	87.46	35.69	2.450546	0.071168
C2	0.62	3.22	1.64	1.963415	0.378049
C3	0.13	0.17	0.15	1.133333	0.866667
i-C4	1.26	0.74	1.06	0.698113	1.188679
n-C4	0.18	0.07	0.14	0.5	1.285714
i-C5	2.04	0.33	1.37	0.240876	1.489051
n-C5	0.11	0.01	0.07	0.142857	1.571429
C6	3.76	0.26	2.39	0.108787	1.573222
C7+	88.81	0.39	54.29	0.007184	1.635845
	100	100	100		



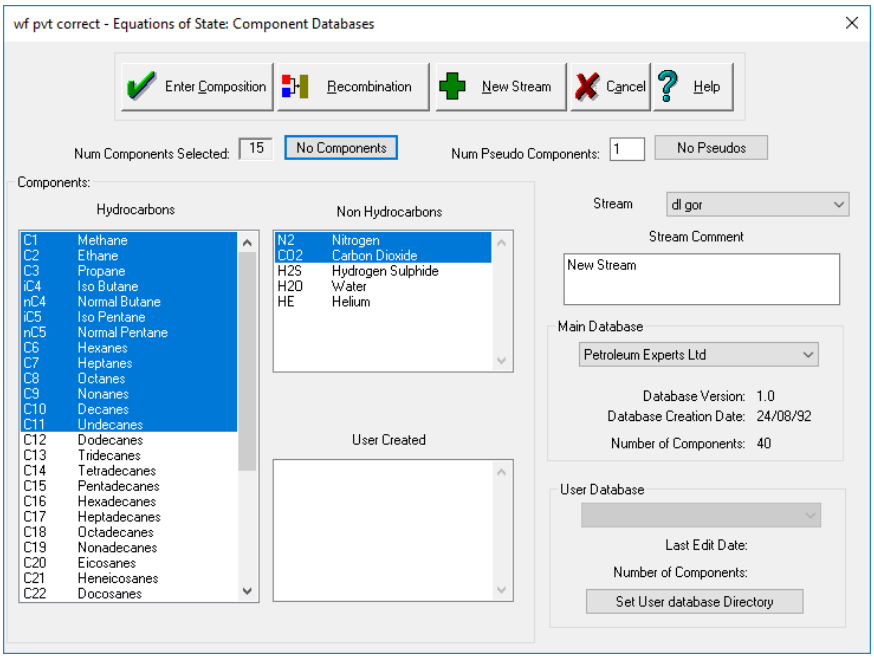
STEP 3: SET UP PVTP

1. In the PVTP software, click **File**, then **New**.
2. Click **Options**, then **Units**, and change the units you need: density from kg/m³ to g/cc, and pressure to psia.



3. Click **Data**. You can enter the data either as individual components or as a list.

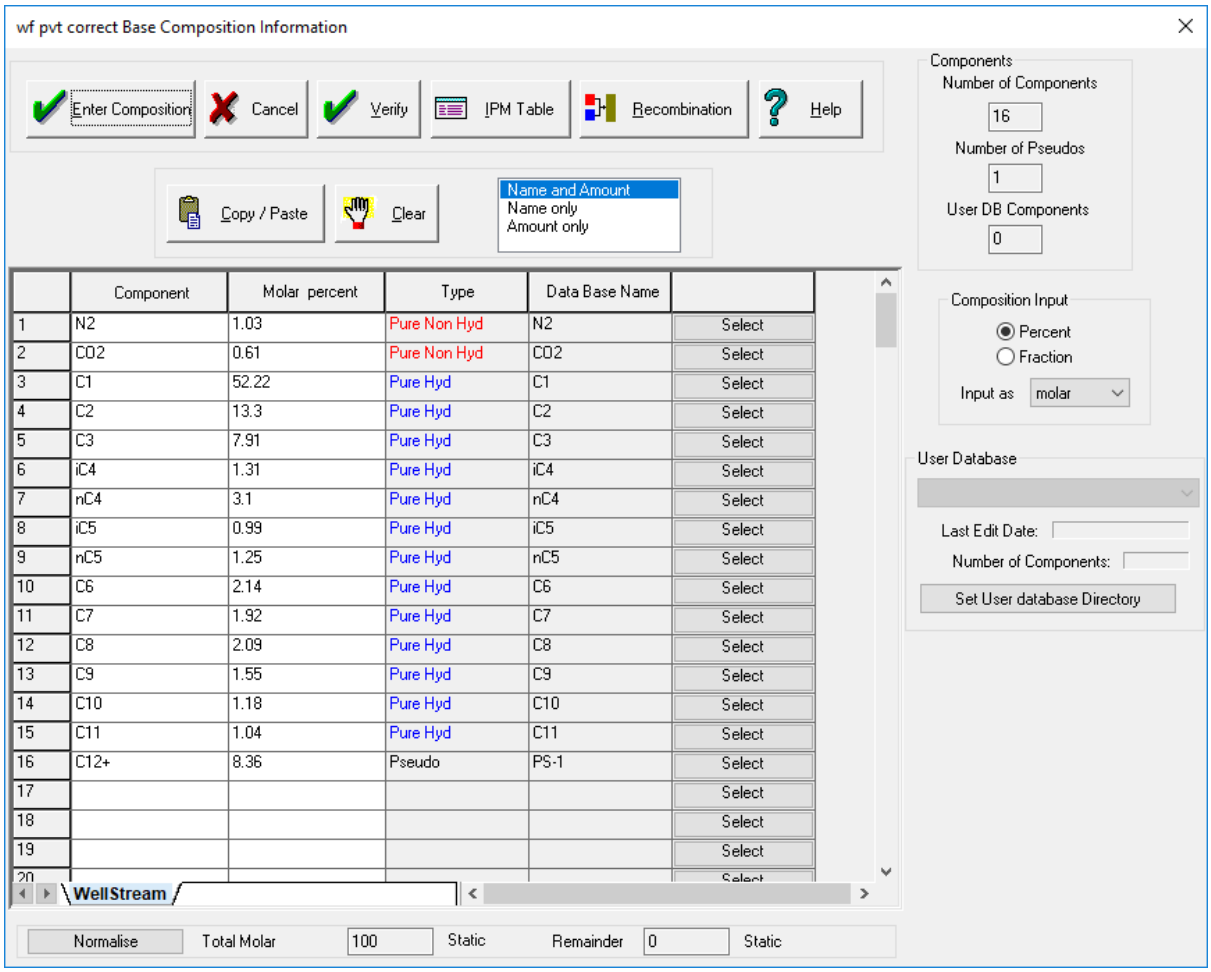
Individual components: you will see the components PVTP accepts. Choose the ones that correspond to your data, click **Enter Composition** and input the values. A page then opens where you input the molecular weight of the pseudo-component, the reservoir temperature, the reservoir (initial) pressure and the reference temperature. Then click **Pseudo Properties**.

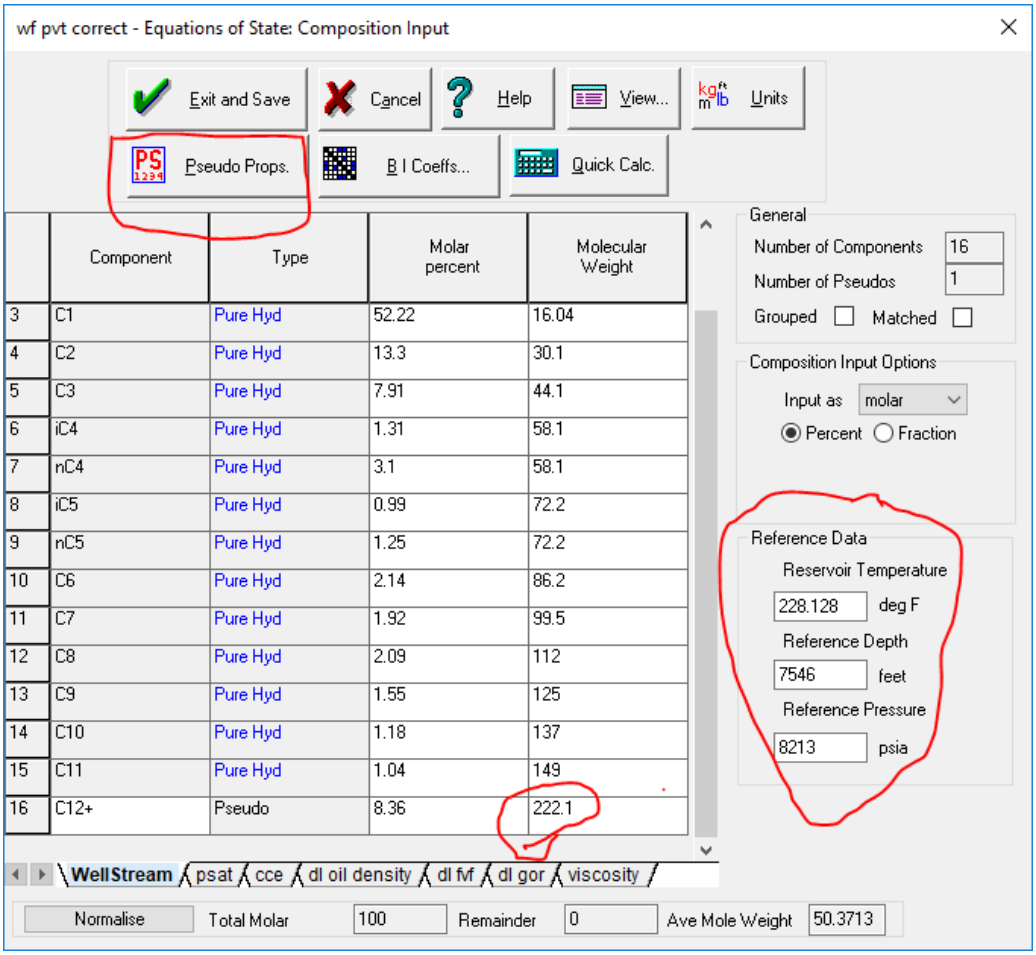


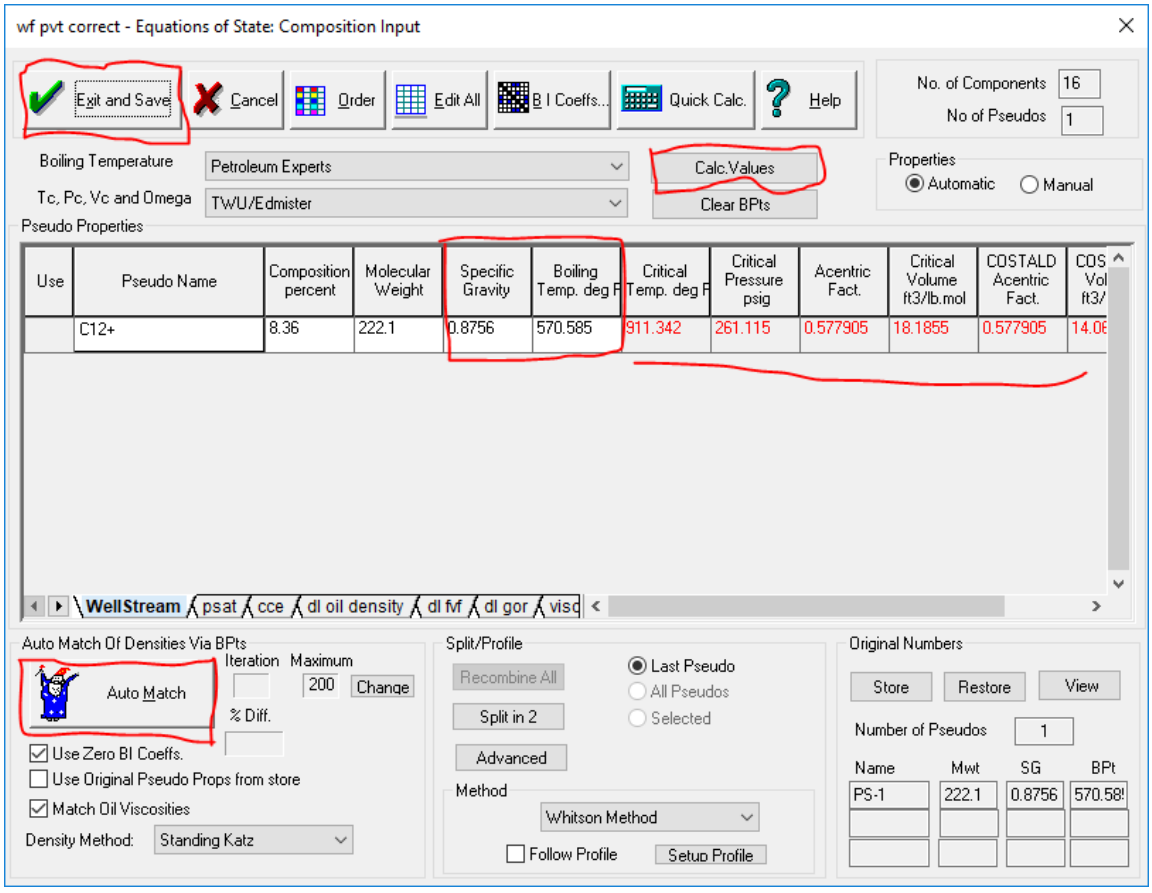
List: click **Data**, then **Enter Components as List**, and paste your components directly from your Excel sheet. Once pasted, click **Verify**. The software identifies the hydrocarbons, the non-hydrocarbons and the pseudo-component (the last component, with a plus sign after it, e.g. C7+, C12+). Then click **Enter Composition**. A page opens where you input the molecular weight of the pseudo-component, the reservoir temperature, the reservoir (initial) pressure and the reference temperature. Then click **Pseudo Properties**.

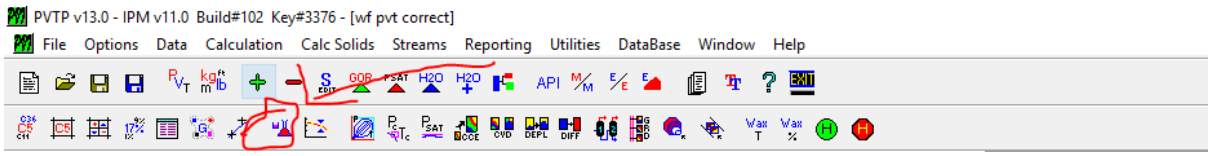
Once you click Pseudo Properties, a new screen appears. Input the specific gravity and boiling point (if you have them), click **Calc Values**, then **Automatch**. This calculates the properties of your pseudo-component. Then click **Exit** and save.

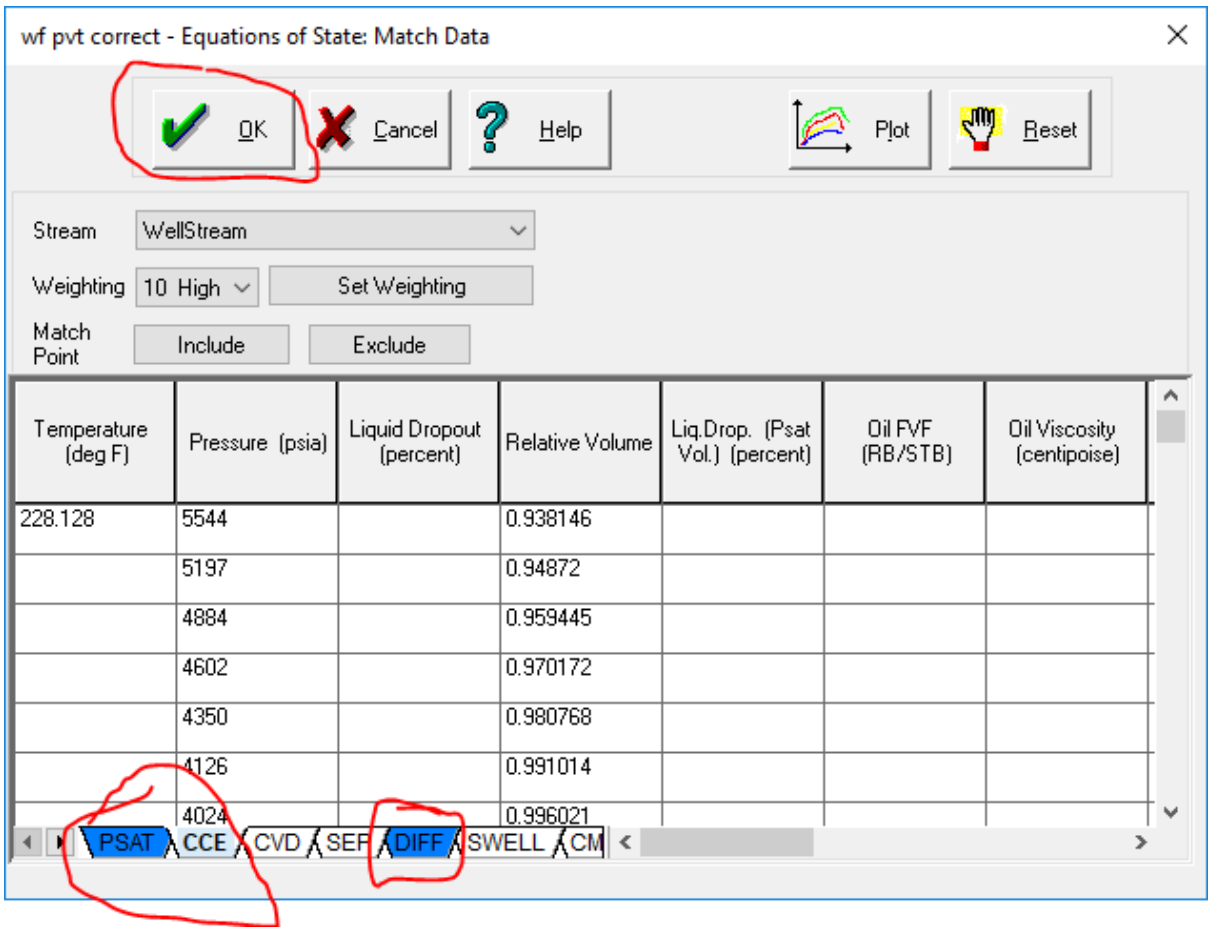
The last stage is to input your experiment data. Click the purple beaker icon and you will see the various experiments: Psat, CCE, DL, CVD, etc. Psat, CCE and DL are used for this example. Take the information from your extracted data and put it into the matching columns.




wf pvt correct - Equations of State: Composition Input
Exit and Save
Cancel
Help
View...
Units
Pseudo Props.
BI Coeffs...
Quick Calc.
Component
Type
Molar percent
Molecular Weight
3 C1 Pure Hyd 52.22 16.04
4 C2 Pure Hyd 13.3 30.1
5 C3 Pure Hyd 7.91 44.1
6 iC4 Pure Hyd 1.31 58.1
7 nC4 Pure Hyd 3.1 58.1
8 iC5 Pure Hyd 0.99 72.2
9 nC5 Pure Hyd 1.25 72.2
10 C6 Pure Hyd 2.14 86.2
11 C7 Pure Hyd 1.92 99.5
12 C8 Pure Hyd 2.09 112
13 C9 Pure Hyd 1.55 125
14 C10 Pure Hyd 1.18 137
15 C11 Pure Hyd 1.04 149
16 C12+ Pseudo 8.36 222.1
General
Number of Components 16
Number of Pseudos 1
Grouped
Matched
Composition Input Options
Input as molar
Percent
Fraction
Reference Data
Reservoir Temperature
228.128 deg F
Reference Depth
7546 feet
Reference Pressure
8213 psia
WellStream
psat
cce
dl oil density
dl fvf
dl gor
viscosity
Normalise
Total Molar 100
Remainder 0
Ave Mole Weight 50.3713


wf pvt correct - Equations of State: Composition Input
Exit and Save
Cancel
Order
Edit All
BI Coeffs...
Quick Calc.
Help
No. of Components 16
No of Pseudos 1
Boiling Temperature
Petroleum Experts
Tc, Pc, Vc and Omega
TWU/Edmister
Calc.Values
Clear BPts
Properties
Automatic
Manual
Pseudo Properties
Use
Pseudo Name
Composition percent
Molecular Weight
Specific Gravity
Boiling Temp. deg F
Critical Temp. deg F
Critical Pressure psig
Acentric Fact.
Critical Volume ft3/lb.mol
COSTALD Acentric Fact.
C12+ 8.36 222.1 0.8756 570.585 911.342 261.115 0.577905 18.1855 0.577905
WellStream
psat
cce
dl oil density
dl fvf
dl gor
Auto Match Of Densities Via BPts
Auto Match
Iteration
Maximum
200
Change
% Diff.
Use Zero BI Coeffs.
Use Original Pseudo Props from store
Match Oil Viscosities
Density Method: Standing Katz
Split/Profile
Recombine All
Split in 2
Advanced
Last Pseudo
All Pseudos
Selected
Method
Whitson Method
Follow Profile
Setup Profile
Original Numbers
Store
Restore
View
Number of Pseudos 1
Name Mwt SG BPt
PS-1 222.1 0.8756 570.58

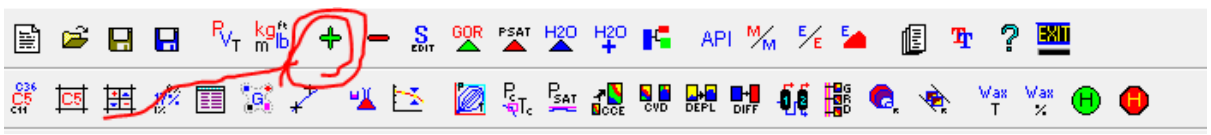


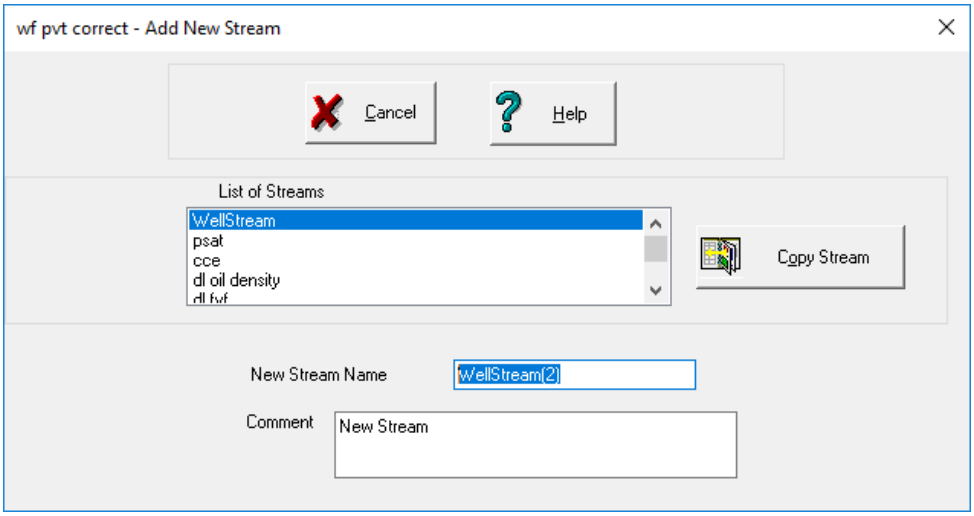


STEP 4: PSAT

The first thing we do is copy our existing stream. Creating a stream is like creating a particular fluid that you want the software to work with. When we start manipulating the stream, we are changing things about that fluid. To preserve our fluid, we copy the stream, so if anything goes wrong we still have a reference for a good fluid.

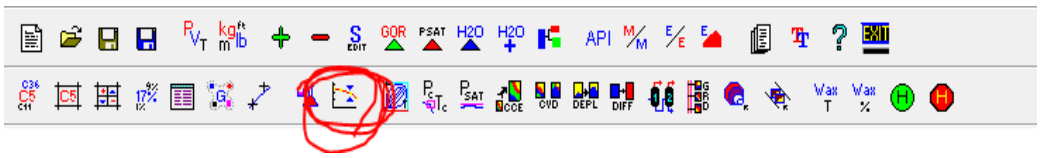
Click the **+** button above, then **Copy Existing Stream**, and name your new stream. We will name ours Psat.



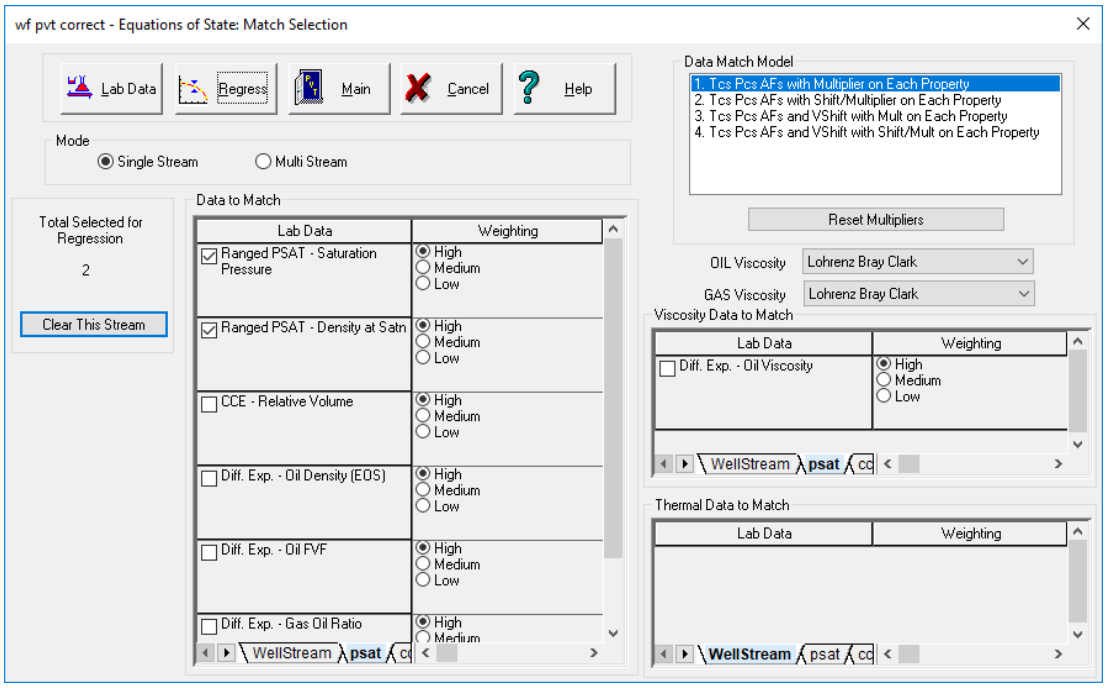


Next is regression. When we regress, we tell the EOS to tune the stream towards certain properties, in this case the saturation pressure. The lower the error, the better the regression.

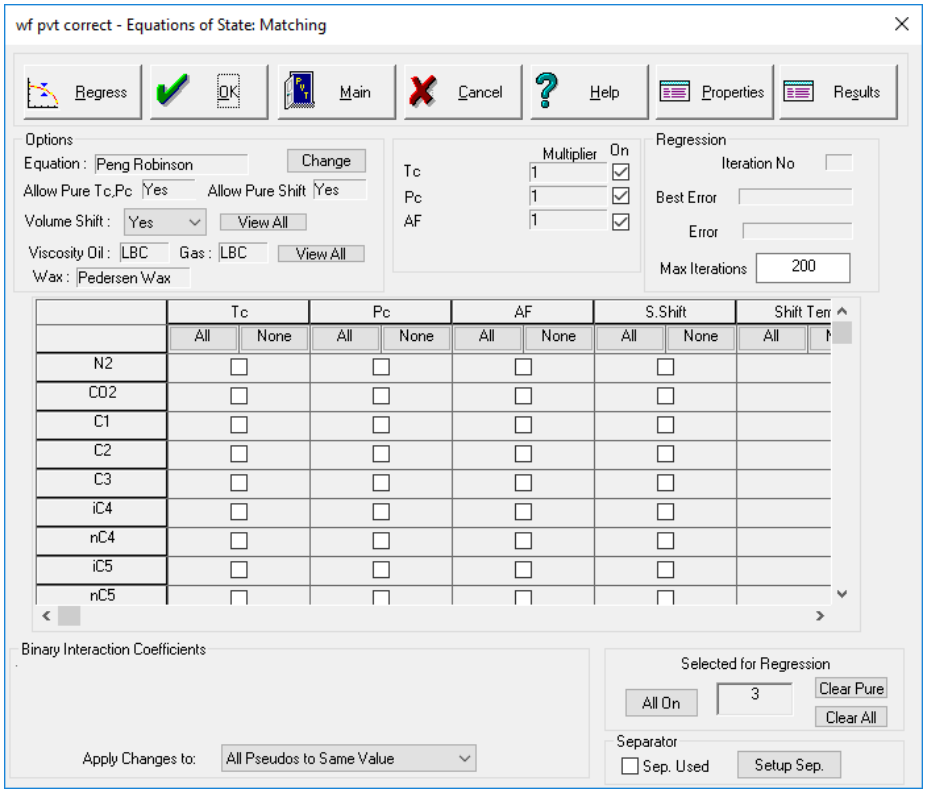
1. Click the icon that looks like a scatter graph, then toggle **Saturation Pressure (Bubble Point)** and **Density at Saturation Pressure**.



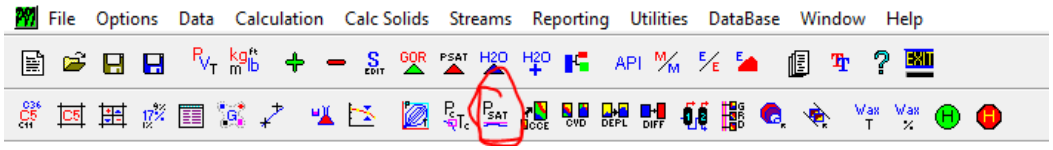
2. Under the Lab Data column, click **Ranged Psat**. Set the formula for oil and gas viscosity to any one you like and click **Regress**.

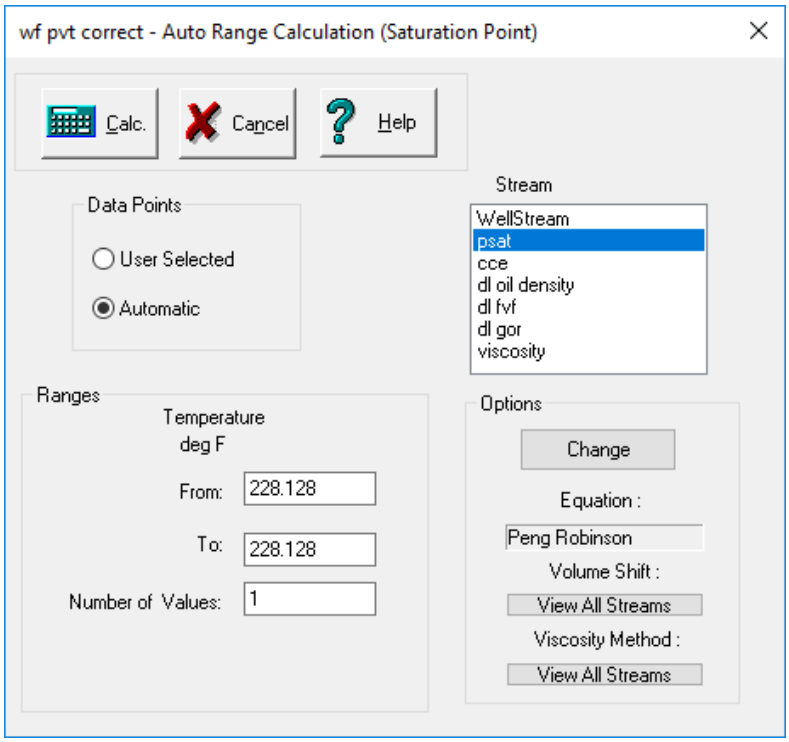


3. Another screen pops up showing the parameters you can adjust to tune the regression. Click **All On**, then **Clear Pure**, then **Regress**.

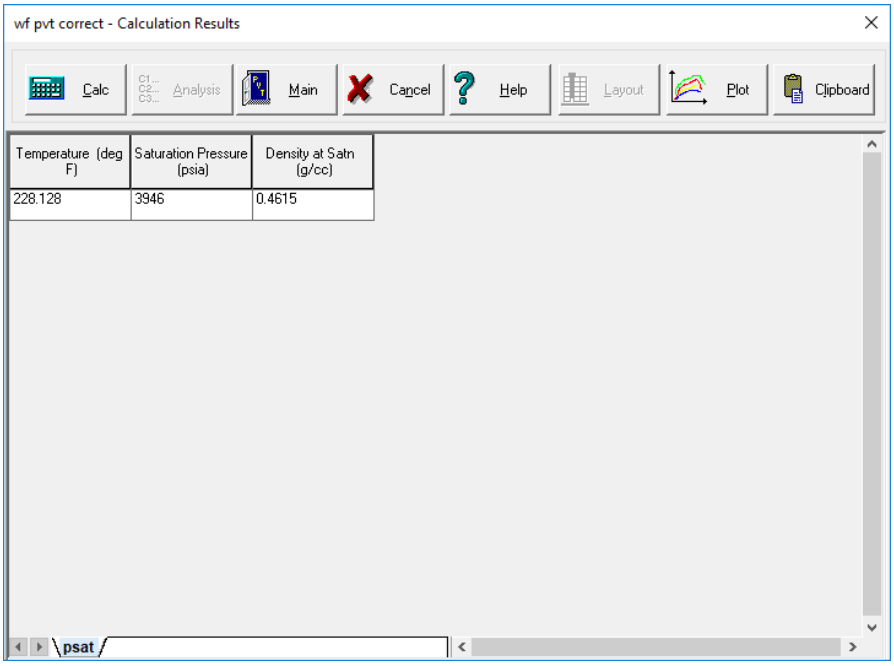


4. Go to the ribbon and click the **Psat** icon. Insert the temperature (make sure it is on Automatic first), set the number of values to one, and click **Calculate**. A new screen appears; click **Calculate** again.





Compare the calculated Psat with the lab value. A common rule of thumb I use is that a result within 3% of the lab value is a good match. If it is not, you still have some regression to do, or you need to change one or two parameters to get a better match. Once the match is good, click **Main** to save your work.

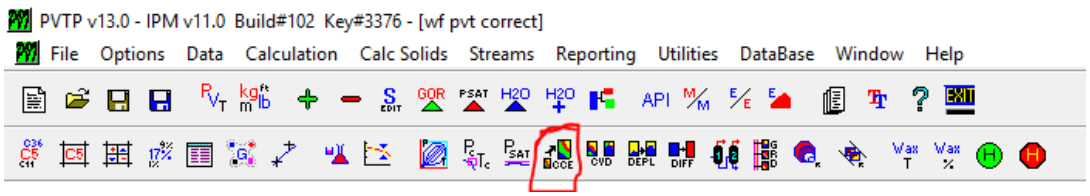


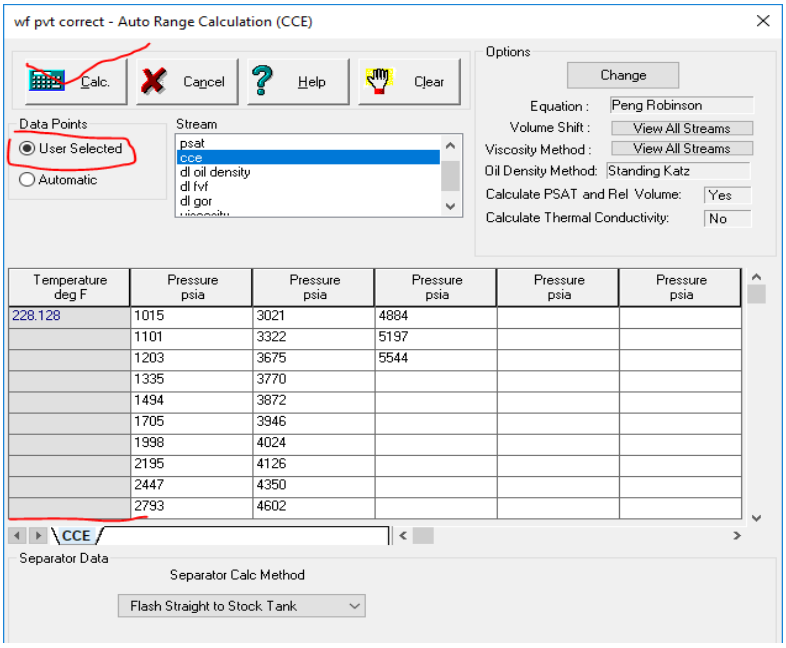
STEP 5: CALCULATE CCE AND DL

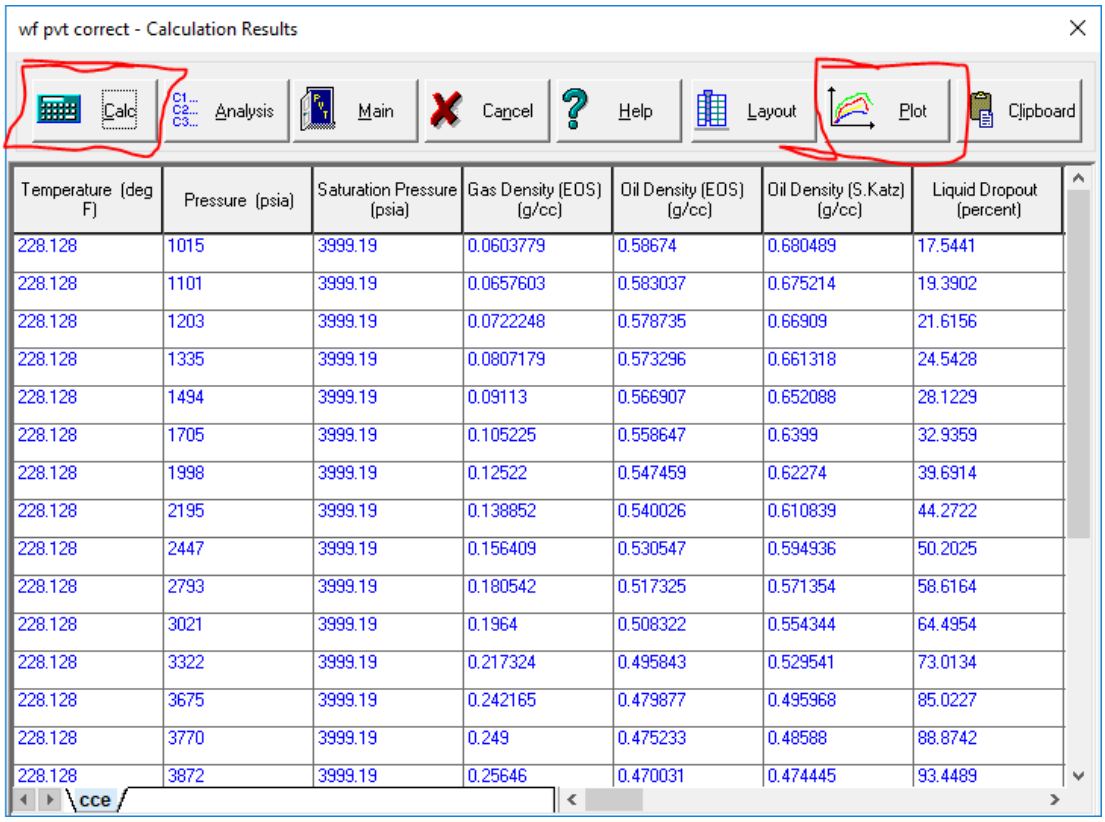
To calculate the other two experiments, we follow similar steps.

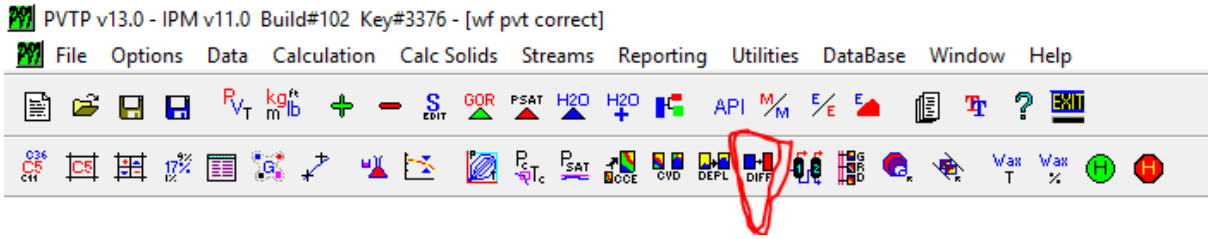
CCE: create a new stream and regress, but this time toggle **CCE**. Then click the **CCE** icon and input the temperature and pressure values. When pasting the pressure values, stop your list where the highlighted blue box stops, then move on to the next set of values until you are done pasting. Make sure it is set to **User Selected**, click **Calculate**, then **Plot**. If your curve matches the lab points, the model is good. If not, adjust some regression parameters, then click **Main** to save.

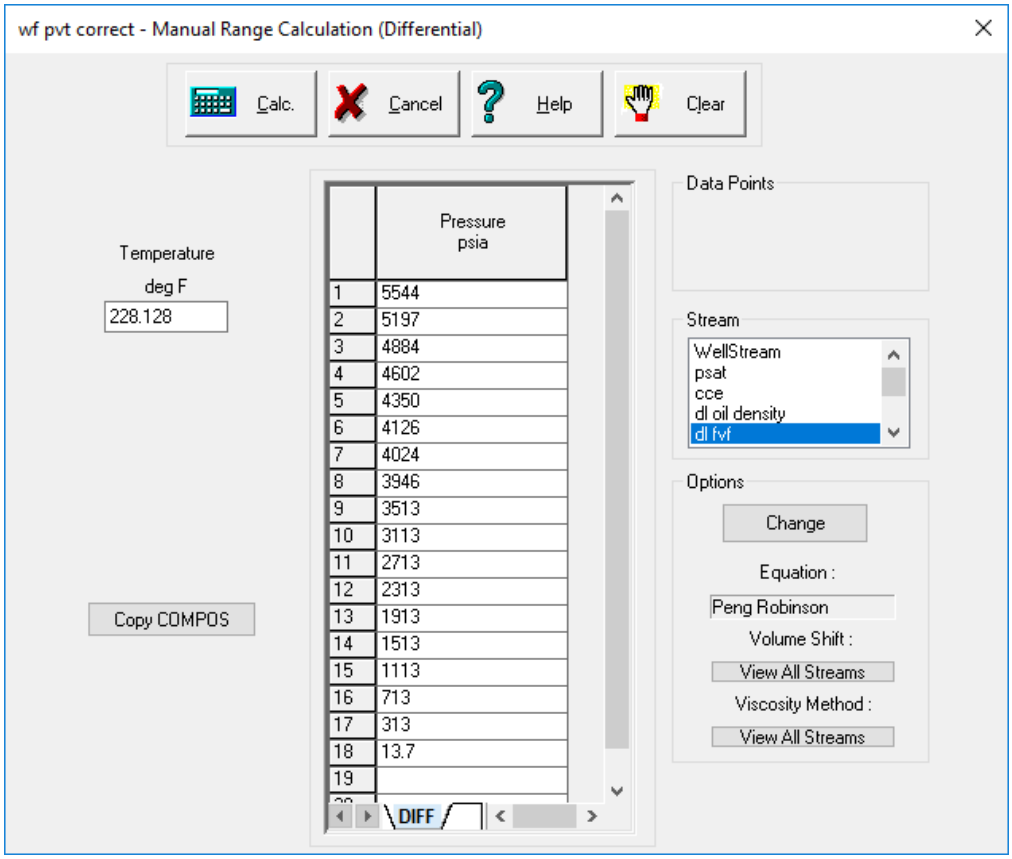
DL: under differential liberation there are three main properties we look at: oil density, oil formation volume factor (Bo) and gas-oil ratio (GOR). For this example, take each one under a new stream and regress for the property you are trying to match. Then click the **DL** icon, insert the temperature and pressure values, click **Calculate**, then **Plot**. Go to **Variables** to change the chart between Bo, GOR and density. Finally, click **Main** to save.

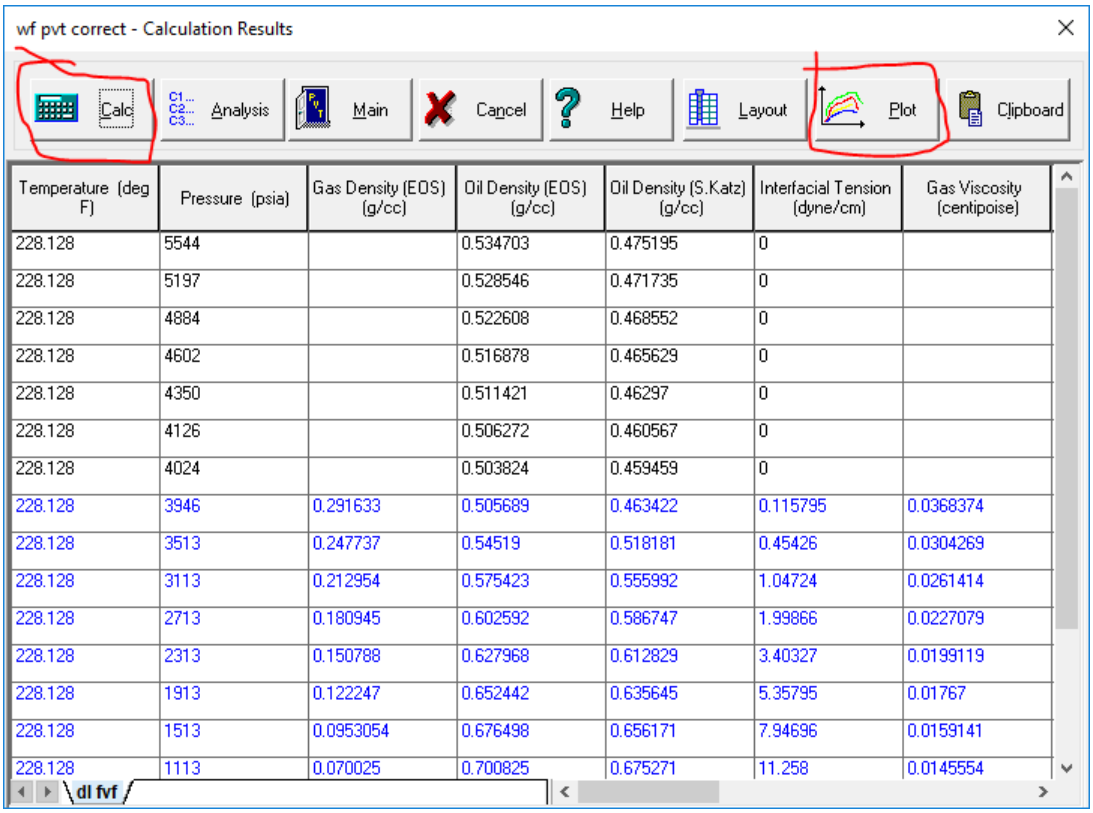



wf pvt correct - Auto Range Calculation (CCE)
Calc.
Cancel
Help
Clear
Options
Change
Equation : Peng Robinson
Volume Shift : View All Streams
Viscosity Method : View All Streams
Oil Density Method: Standing Katz
Calculate PSAT and Rel Volume: Yes
Calculate Thermal Conductivity: No
Data Points
User Selected
Automatic
Stream
psat
cce
dl oil density
dl fvf
dl gor
Temperature deg F
Pressure psia
228.128
1015
3021
4884
1101
3322
5197
1203
3675
5544
1335
3770
1494
3872
1705
3946
1998
4024
2195
4126
2447
4350
2793
4602
CCE
Separator Data
Separator Calc Method
Flash Straight to Stock Tank


wf pvt correct - Calculation Results
Calc
Analysis
Main
Cancel
Help
Layout
Plot
Clipboard
Temperature (deg F)
Pressure (psia)
Saturation Pressure (psia)
Gas Density (EOS) (g/cc)
Oil Density (EOS) (g/cc)
Oil Density (S.Katz) (g/cc)
Liquid Dropout (percent)
228.128 1015 3999.19 0.0603779 0.58674 0.680489 17.5441
228.128 1101 3999.19 0.0657603 0.583037 0.675214 19.3902
228.128 1203 3999.19 0.0722248 0.578735 0.66909 21.6156
228.128 1335 3999.19 0.0807179 0.573296 0.661318 24.5428
228.128 1494 3999.19 0.09113 0.566907 0.652088 28.1229
228.128 1705 3999.19 0.105225 0.558647 0.6399 32.9359
228.128 1998 3999.19 0.12522 0.547459 0.62274 39.6914
228.128 2195 3999.19 0.138852 0.540026 0.610839 44.2722
228.128 2447 3999.19 0.156409 0.530547 0.594936 50.2025
228.128 2793 3999.19 0.180542 0.517325 0.571354 58.6164
228.128 3021 3999.19 0.1964 0.508322 0.554344 64.4954
228.128 3322 3999.19 0.217324 0.495843 0.529541 73.0134
228.128 3675 3999.19 0.242165 0.479877 0.495968 85.0227
228.128 3770 3999.19 0.249 0.475233 0.48588 88.8742
228.128 3872 3999.19 0.25646 0.470031 0.474445 93.4489
cce


PVTP v13.0 - IPM v11.0 Build#102 Key#3376 - [wf pvt correct]
File Options Data Calculation Calc Solids Streams Reporting Utilities DataBase Window Help
EXIT





Temperature (deg F)	Pressure (psia)	Gas Density (EOS) (g/cc)	Oil Density (EOS) (g/cc)	Oil Density (S.Katz) (g/cc)	Interfacial Tension (dyne/cm)	Gas Viscosity (centipoise)
228.128	5544		0.534703	0.475195	0	
228.128	5197		0.528546	0.471735	0	
228.128	4884		0.522608	0.468552	0	
228.128	4602		0.516878	0.465629	0	
228.128	4350		0.511421	0.46297	0	
228.128	4126		0.506272	0.460567	0	
228.128	4024		0.503824	0.459459	0	
228.128	3946	0.291633	0.505689	0.463422	0.115795	0.0368374
228.128	3513	0.247737	0.54519	0.518181	0.45426	0.0304269
228.128	3113	0.212954	0.575423	0.555992	1.04724	0.0261414
228.128	2713	0.180945	0.602592	0.586747	1.99866	0.0227079
228.128	2313	0.150788	0.627968	0.612829	3.40327	0.0199119
228.128	1913	0.122247	0.652442	0.635645	5.35795	0.01767
228.128	1513	0.0953054	0.676498	0.656171	7.94696	0.0159141
228.128	1113	0.070025	0.700825	0.675271	11.258	0.0145554

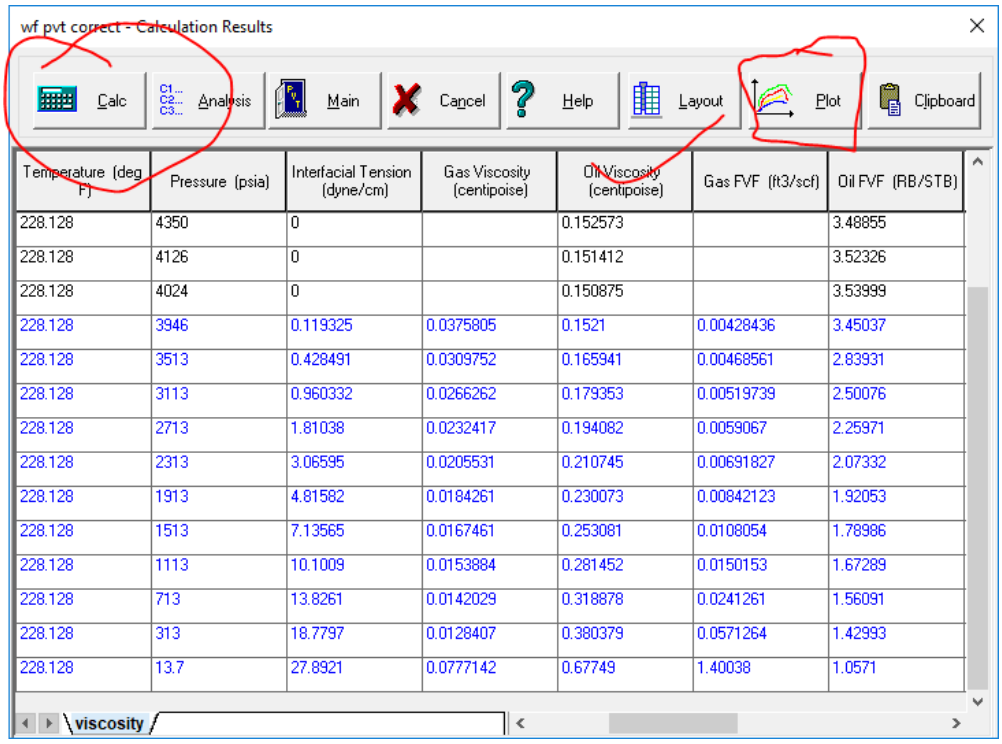
STEP 6: CALCULATE VISCOSITY

Viscosity is a bit tricky, so it is regressed and calculated separately. Regress your viscosity values first. For most of the viscosity formulas you don't need to click All On or Clear Pure; just click **Regress**.

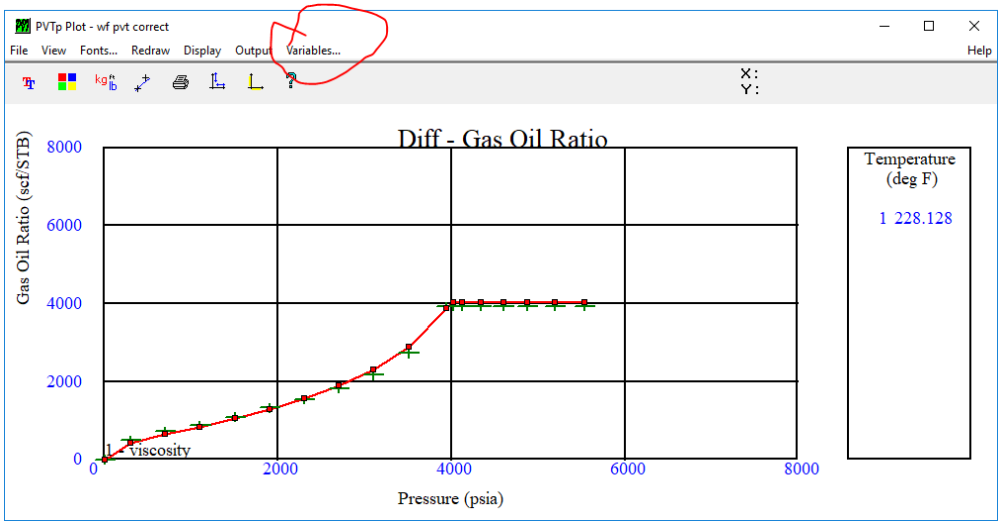
To calculate, click the **DL** icon, then **Calculate** and **Plot**. Go to **Variables** and change the variable to **Oil Viscosity**. This is the same way you switch between Bo, oil density and GOR: once you have plotted, just go to Variables and change the variable.

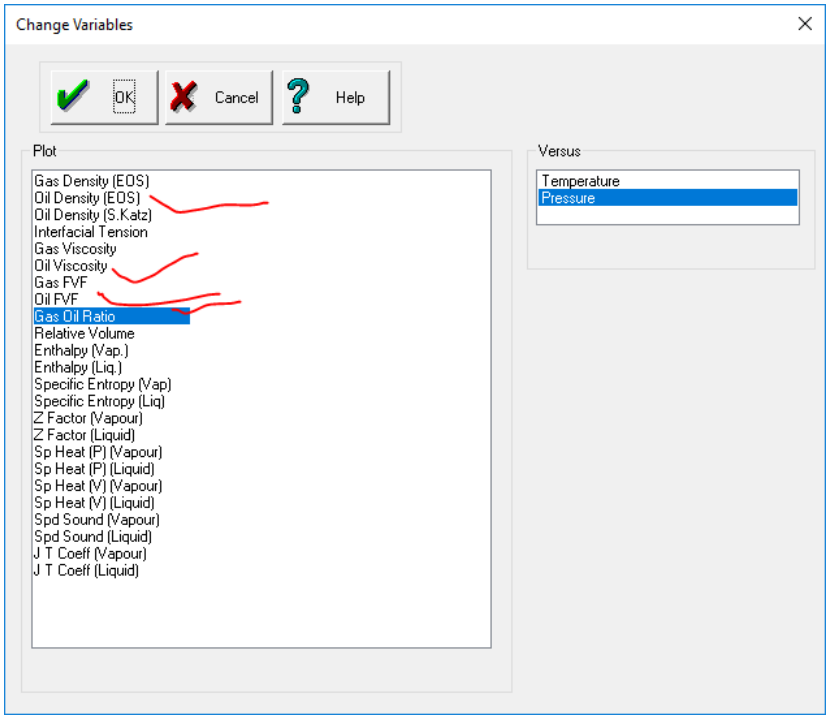
If the viscosity does not match, try a different viscosity formula, such as LBC (Lohrenz-Bray-Clark), Pedersen et al., Zhou et al. or Beggs and Beal, and see which gives the lowest error.

Once you are done with this workflow and all your graphs match, congratulations! You have now modelled the fluid. You can use it for material balance or well modelling. Hurray!!!



Temperature (deg F)	Pressure (psia)	Interfacial Tension (dyne/cm)	Gas Viscosity (centipoise)	Oil Viscosity (centipoise)	Gas FVF (ft3/scf)	Oil FVF (RB/STB)
228.128	4350	0		0.152573		3.48855
228.128	4126	0		0.151412		3.52326
228.128	4024	0		0.150875		3.53999
228.128	3946	0.119325	0.0375805	0.1521	0.00428436	3.45037
228.128	3513	0.428491	0.0309752	0.165941	0.00468561	2.83931
228.128	3113	0.960332	0.0266262	0.179353	0.00519739	2.50076
228.128	2713	1.81038	0.0232417	0.194082	0.0059067	2.25971
228.128	2313	3.06595	0.0205531	0.210745	0.00691827	2.07332
228.128	1913	4.81582	0.0184261	0.230073	0.00842123	1.92053
228.128	1513	7.13565	0.0167461	0.253081	0.0108054	1.78986
228.128	1113	10.1009	0.0153884	0.281452	0.0150153	1.67289
228.128	713	13.8261	0.0142029	0.318878	0.0241261	1.56091
228.128	313	18.7797	0.0128407	0.380379	0.0571264	1.42993
228.128	13.7	27.8921	0.0777142	0.67749	1.40038	1.0571





EXPECTED RESULTS

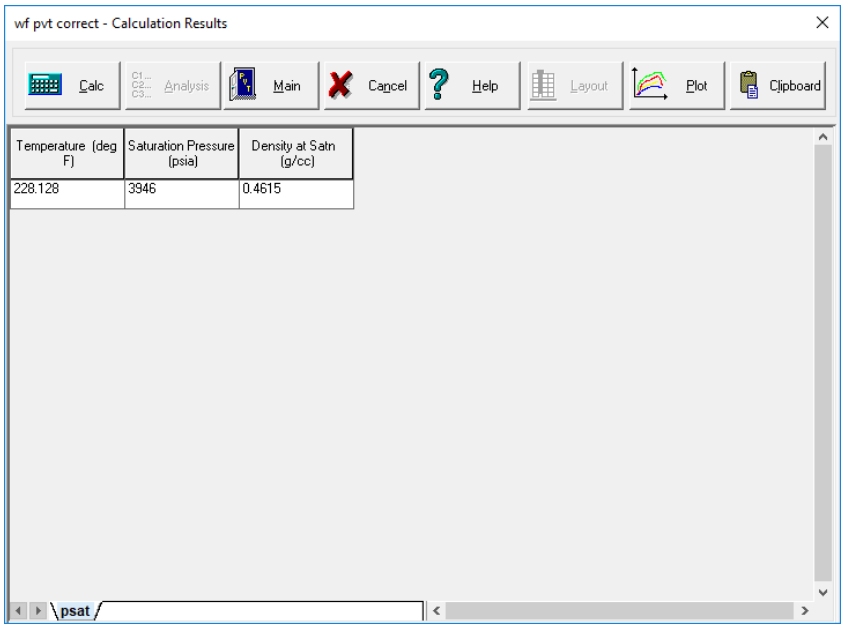
For every test you run, you should get a graph that matches the reservoir fluid properties you input. On the chart you will see a red line and some green points. The green points are the lab data and show the true pattern of the reservoir fluid. The red line is the model you are building. If the red line follows the pattern of the green points, sitting on them, your model matches the fluid.

Once you confirm a match, go to the calculated values, copy them out and compare them with your lab values. Calculate the absolute error for each point:

$$\textbf{Absolute error (\%)} = |\textbf{Calc} - \textbf{Lab}| \div \textbf{Lab} \times 100$$

Then calculate the average absolute error (AAE). If the AAE is between 0 and 5%, it is a good match.

Psat



CCE

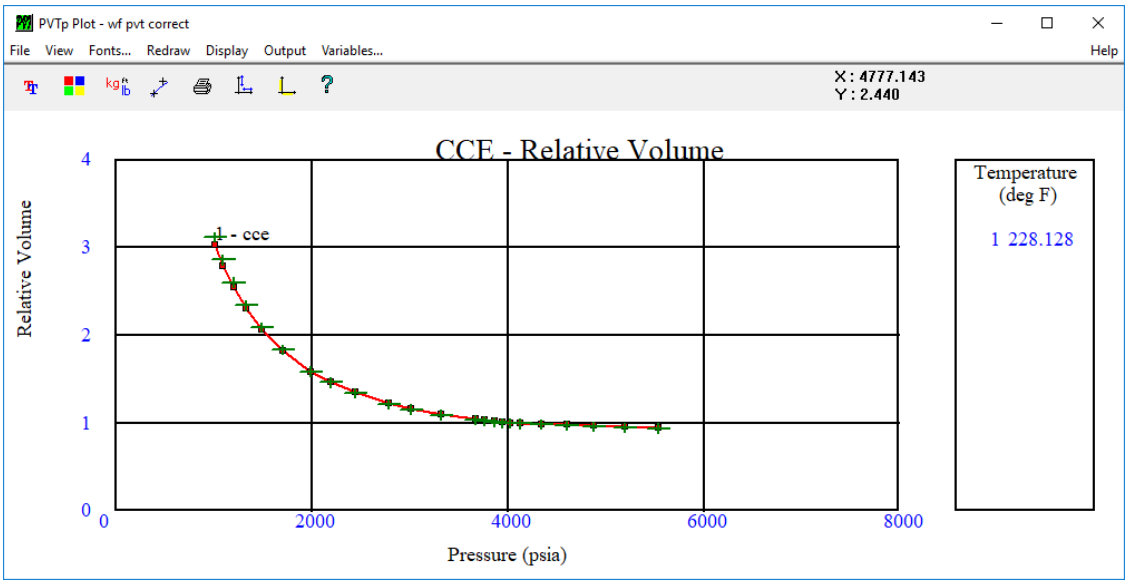


Table 1: CCE relative volume, lab vs calculated

Lab Values	Calc Values	Abs Error (%)
0.938146	0.949285	1.19
0.94872	0.958867	1.07
0.959445	0.968261	0.92
0.970172	0.977471	0.75
0.980768	0.986373	0.57
0.991014	0.994894	0.39
0.996021	0.998984	0.30

Lab Values	Calc Values	Abs Error (%)
1	1.00644	0.64
1.008499	1.01568	0.71
1.021057	1.02917	0.79
1.033707	1.04258	0.86
1.090372	1.10105	0.98
1.154221	1.16504	0.94
1.215586	1.22553	0.82
1.338851	1.34556	0.50
1.461143	1.46366	0.17
1.58419	1.58187	0.15
1.832782	1.81955	0.72
2.08475	2.05925	1.22
2.336109	2.29746	1.65
2.60239	2.54907	2.05
2.857146	2.78928	2.38
3.115874	3.03288	2.66
	AAE (%)	**0.98**

Above the bubble point (while the fluid is undersaturated), the relative volume rises only slightly as pressure drops. This is because liquids are nearly incompressible. Once pressure falls below the bubble point, gas begins to come out of the oil. Gas is far more compressible than liquid, so the volume shoots up sharply. The total volume increases as pressure drops.

The AAE is about 1%. This is good and means our model represents the fluid well and can be used.

DL: Oil Density

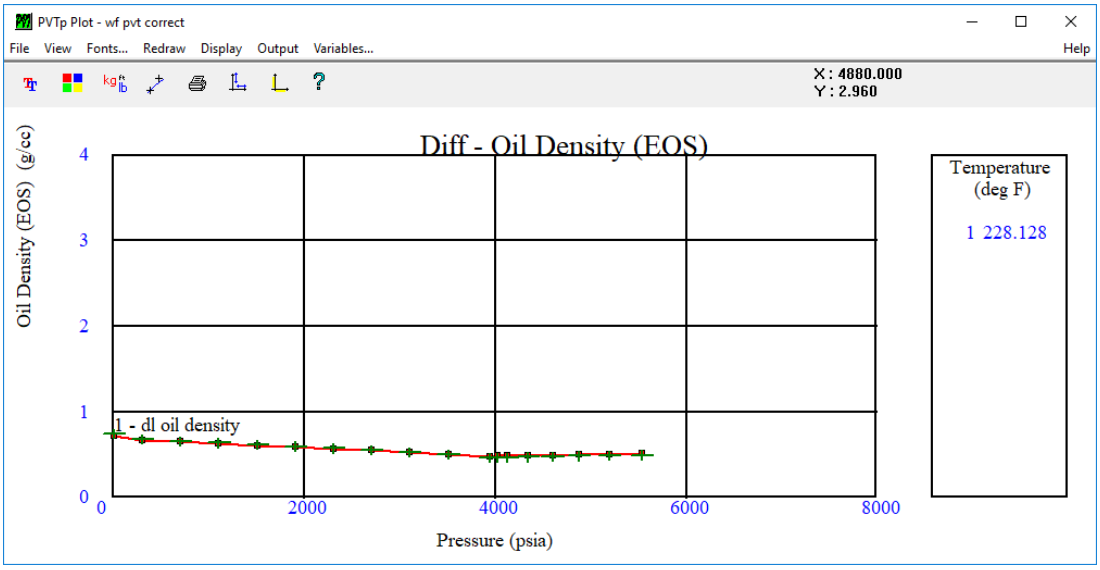


Table 2: Oil density (g/cc), lab vs calculated

Lab Values	Calc Values	Abs Error (%)
0.4919	0.508327	3.34
0.4864	0.502777	3.37
0.481	0.49743	3.42
0.4757	0.492277	3.48
0.4705	0.487377	3.59
0.4657	0.482758	3.66
0.4633	0.480565	3.73
0.4615	0.478848	3.76
0.4981	0.497355	0.15
0.5264	0.522844	0.68
0.5512	0.545629	1.01
0.5733	0.566795	1.13
0.5945	0.587089	1.25
0.6146	0.606892	1.25
0.6369	0.626716	1.60
0.6541	0.647345	1.03
0.6819	0.671854	1.47
0.7427	0.722446	2.73
	AAE (%)	**2.26**

Above the bubble point, oil density decreases slightly as pressure drops, because the oil expands. At the bubble point, the density reaches its minimum. Below the bubble point, density increases as

pressure drops further. This is because the light components escape as gas, leaving behind the heavier components, so the remaining oil becomes denser.

DL: Oil Formation Volume Factor (Bo)

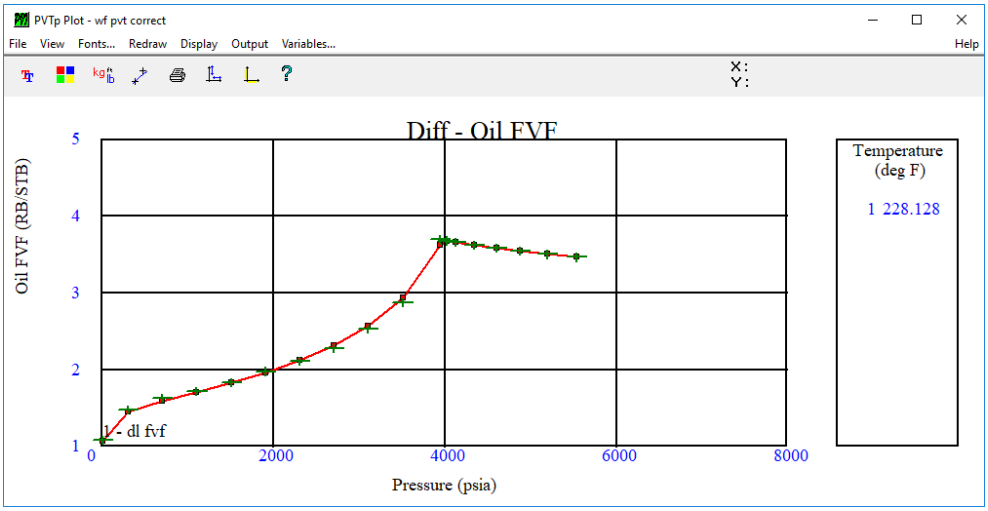


Table 3: Oil formation volume factor, lab vs calculated

Lab Values	Calc Values	Abs Error (%)
3.4632	3.46493	0.05
3.5023	3.50529	0.09
3.5419	3.54512	0.09
3.5815	3.58442	0.08
3.6206	3.62266	0.06
3.6584	3.65951	0.03
3.6769	3.67729	0.01
3.6916	3.61408	2.10
2.8739	2.93637	2.17
2.5265	2.57274	1.83
2.2764	2.31751	1.81
2.1122	2.12178	0.45
1.9743	1.96219	0.61
1.8317	1.8262	0.30
1.7196	1.70478	0.86
1.6249	1.58873	2.23
1.4757	1.45298	1.54
1.0842	1.06415	1.85
	AAE (%)	**0.90**

Above the bubble point, Bo increases slightly as pressure drops, because the oil is expanding. At the bubble point it is at its maximum. Below the bubble point it decreases, because gas is coming out of solution and the oil shrinks, leaving less total oil volume.

DL: Gas-Oil Ratio (GOR)

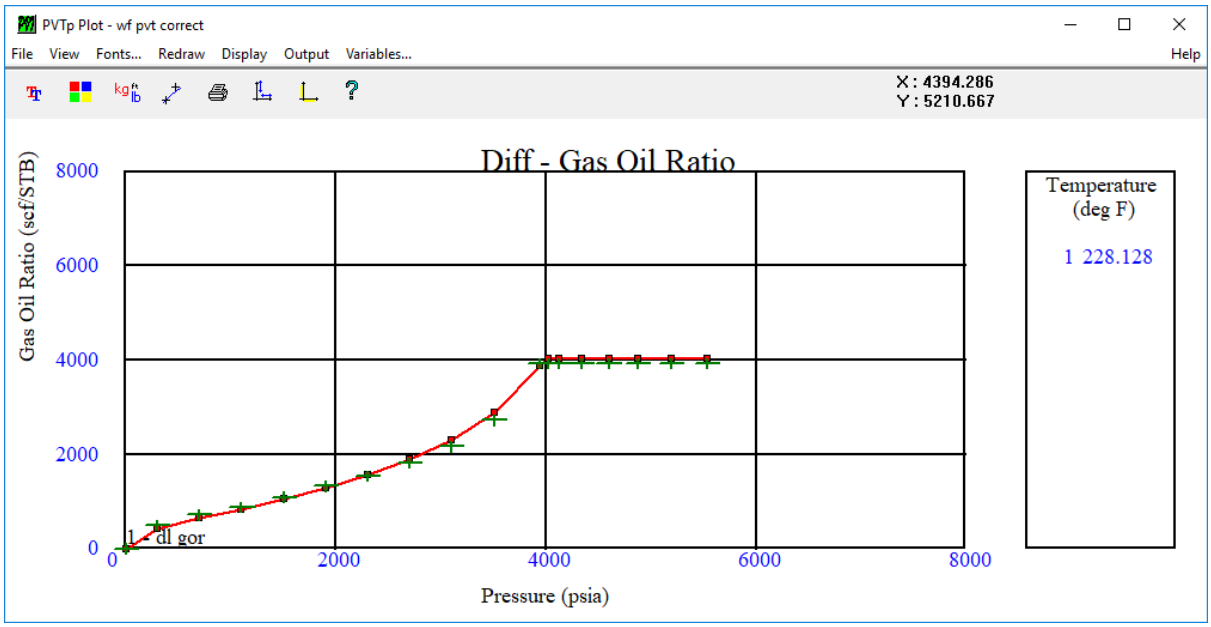


Table 4: Solution GOR (scf/stb), lab vs calculated

Lab Values	Calc Values	Abs Error (%)
3918.59	4015.05	2.46
3918.59	4015.05	2.46
3918.59	4015.05	2.46
3918.59	4015.05	2.46
3918.59	4015.05	2.46
3918.59	4015.05	2.46
3918.59	4015.05	2.46
3918.59	3878.35	1.03
2718.48	2875.56	5.78
2159.22	2300.7	6.55
1816.32	1881.19	3.57
1537.08	1550.72	0.89
1319.4	1277.05	3.21
1085.44	1041.71	4.03
879.39	832.135	5.37
713.09	635.951	10.82
500.45	422.941	15.49
	AAE (%)	**4.35**

Note: the final point, where the lab GOR is 0, is left out because the error calculation divides by zero (#DIV/0!).

Above the bubble point, all the gas is still dissolved in the oil and none escapes, so the GOR stays constant. Once pressure falls below the bubble point, gas begins to escape and the curve turns downward.

Viscosity

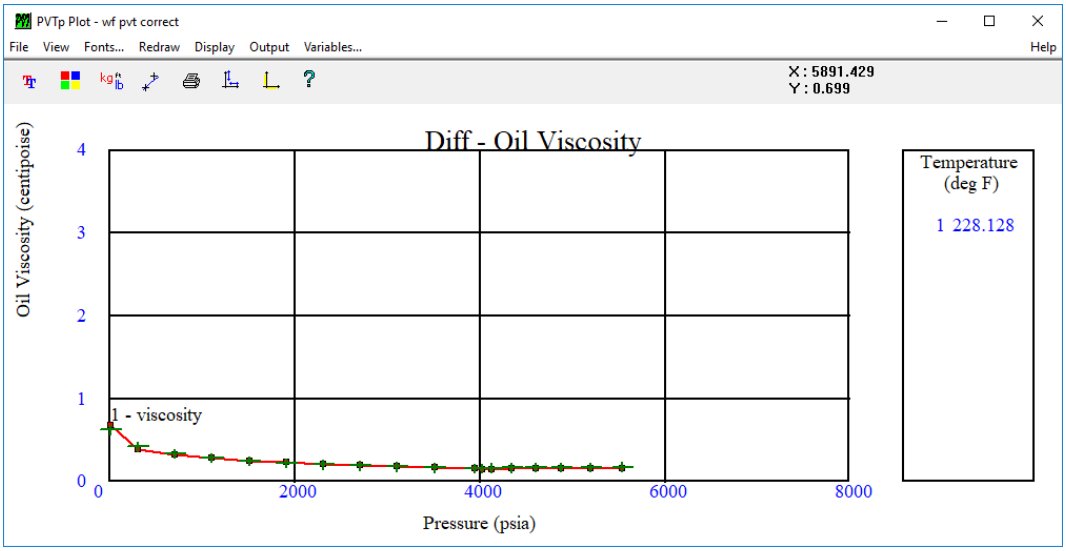


Table 5: Oil viscosity (cP), lab vs calculated

Lab Values	Calc Values	Abs Error (%)
0.17531	0.15843	9.63
0.17184	0.156779	8.76
0.16871	0.155256	7.97
0.16589	0.153853	7.26
0.16316	0.152573	6.49
0.16017	0.151412	5.47
0.1584	0.150875	4.75
0.157	0.1521	3.12
0.163	0.165941	1.80
0.178	0.179353	0.76
0.19	0.194082	2.15
0.204	0.210745	3.31
0.223	0.230073	3.17
0.25	0.253081	1.23
0.288	0.281452	2.27
0.334	0.318878	4.53

Lab Values	Calc Values	Abs Error (%)
0.42	0.380379	9.43
0.624	0.67749	8.57
	AAE (%)	**5.04**

Above the bubble point, viscosity decreases slightly as pressure drops. High pressure packs the liquid molecules closely together, which increases resistance to flow. Reducing the pressure lets them move more freely. Viscosity reaches its minimum at the bubble point. Below the bubble point, it increases sharply as pressure drops further. Light molecules act as "lubricants" in the oil. As they escape into the gas phase, the remaining heavy molecules make the oil thicker and stickier.

The AAE here is slightly above 5%. This is still acceptable because viscosity is generally harder to match than volumetric properties.

HOW TO TUNE PVTP

Sometimes, when you put in your parameters, you don't get an exact replication of the fluid you have. This means the software has gone astray in its calculations and you need to bring it back. You do this through tuning.

You tune using weights, regression formulas and by changing parameters.

Weights: in PVTP, weighting (scored 0–10) acts as a steering wheel during regression. It tells the software which lab data points are most important to match when tuning its EOS model. No mathematical model can pass through every single lab point perfectly, so weights assign an "error penalty":

- **0** tells PVTP to ignore a bad data point entirely.
- **1** is the default baseline for standard points.
- **5–8** forces the curve closer to highly accurate lab values.
- **10** acts as a strict anchor for non-negotiable physical points, like the bubble-point pressure.

Weights work relatively, so giving everything a 10 is the same as giving everything a 1. The best strategy is to keep most points at 1 and assign high weights only to critical anchors, so the model fits real-world behaviour without distorting the overall curve.

Formulas and parameters: try a different formula or adjust different regression parameters, and see which gives a lower error for the particular property you are matching. For viscosity, the choice of viscosity correlation matters. For CCE and DL volumes, it is the EOS and regression parameters that matter.

Check your data: sometimes the issue comes from the data itself. Look carefully for points that are off, odd or don't follow the trend, because they could be errors. If you notice any funny points, exclude them from the calculation and see if the graph improves. You can even remove them from the data completely.

So there you have it, kids. Thank you for listening. I hope this helped you a little in your PVTP endeavours. You can put any questions you have below.